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Professional Summary

An LLM Scientist with a **Ph.D. in Computer Science** specializing in **NLP and Large Language Models**, and **2+ years of industry** experience at Amazon. Expert in **post-training (GRPO/DPO), agentic systems, RAG, context management, and personalization**. Author of **20+ publications (8 first-authored)** in top venues (NeurIPS, ACL, EMNLP, NAACL etc.) and recipient of a **US Patent**. Lead author of **pioneering works in early multi-agent systems and hyper-attention based tabular LLMs**. Delivered significant production gains at Amazon Rufus, including improving faithfulness from **67% to 90%** and personalization helpfulness from **84% to 93%**.

Experience

Amazon 2024.1 – Present
Applied Scientist (full-time) *Seattle, WA & Palo Alto/Santa Clara, CA*

Led post-training research and development for customer-facing LLM agent systems, focusing on context management, personalization, and response faithfulness.

- **At Amazon Rufus:** Co-led the context management and RAG workstream,
  - reduced hallucinations and improved model faithfulness from **67% to 90%** through synthetic IFT/DPO data;
  - improved personalization helpfulness from **84% to 93%** with verifiable rubrics and GRPO;
  - extended the context window from **8K to 128K** through continual pre-training with ABF and YaRN;
  - and increased multi-turn context carryover from **61% to 73%** using DPO with live-traffic failure patterns as negatives.
- **At Amazon Customer Service:** Owned the end-to-end LLM data flywheel for the team's multi-agent system, from post-training data generation and verification to training and weblab evaluation, establishing automated quality gates and repeatable training→measurement loops that impacted all workstreams; for example, **reduced training time by 4× while delivering ∼5% performance gains** in response generation quality through optimization of GRPO training data.
- **Intern Mentorship:** Mentored two interns on LLM research projects in long-context modeling and multi-turn improvement with verifiable GRPO, guiding ideation, experimentation, evaluation, and writing, resulting in one published paper and two papers under review.

Amazon Web Service (AWS) 2023.5 – 2023.8
Applied Scientist (Intern) *Santa Clara, CA*

- Researched LLM-based multi-agent systems for complex problem solving.
- Developed a **collaborative multi-agent** prompting framework that achieved state-of-the-art performance on complex science benchmarks in both zero-shot and few-shot settings.
- Published the work in Findings of NAACL 2024.

Amazon Web Service (AWS) 2022.6 – 2023.1
Applied Scientist (Intern) *Santa Clara, CA & remote*

- Developed a novel **tabular language modeling** approach that represents tables as hypergraphs to better capture structural dependencies.
- Achieved state-of-the-art performance on tabular representation learning benchmarks.
- Published the work at NeurIPS 2023 as a Spotlight paper **(top 5%)**.

Tencent AI Lab 2021.6 – 2021.8
NLP Researcher (Intern) *remote in College Station, TX*

- Built a comprehensive benchmark for zero-shot knowledge base completion, covering state-of-the-art methods and diverse knowledge sources.

- Resulted in a paper accepted to the ICDM 2022 Knowledge Graph Workshop.

Texas A&M University 2019.9 – 2023.12
Research Assistant and Teaching Assistant *College Station, TX*

- Conducted research in NLP and information extraction, including named entity recognition, event extraction, and fine-grained opinion mining.
- Developed graph-based methods to model non-sequential dependencies among entity mentions, improving domain-specific NER and structured information extraction.
- Built datasets and benchmarks for fine-grained language understanding tasks, including location recognition.
- Served as a teaching assistant for CSCE 636 Deep Learning and CSCE 313 Computer Systems.

National Lab of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences 2018.1 – 2019.7
Research Engineer (Intern & Full-time) *Beijing, China*

- Built information extraction and evaluation systems for financial event extraction and causality detection, bridging NLP research and real-world financial text applications.

Innovation Lab of Global Exchange, State Street 2017.7 – 2018.1
Data Analyst (Intern) *Hangzhou, China*

- Supported data cleaning, analysis, visualization, and database construction for financial applications.

Selected Publications

LLM & Foundation Model Training

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. "HYTREL: Hypergraph-enhanced Tabular Data Representation Learning." **NeurIPS 2023, Spotlight (5%)**, **first work to use hyper-attention for tabular LLMs.**
- Ming Li, **Pei Chen**, Chenguang Wang, et al. "Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning." ACL 2025, Findings.
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. "END: Early Noise Dropping for Efficient and Effective Context Denoising." pre-print.
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. "ItD: Large Language Models Can Teach Themselves Induction through Deduction." ACL-2024, long paper.
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. "Aligning Large Language Models with Implicit Preferences from User-Generated Content." ACL 2025, long paper.

Agent

- **Pei Chen**, Shuai Zhang, Boran Han. "CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving." NAACL 2024, long paper, **pioneering work in demonstrating multi-agent systems for complex LLM reasoning.**
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. "Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training." NAACL 2025, long paper.
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. "Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates." EMNLP 2025, long paper.

RAG & Long-context & Multi-turn & Personalization

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. "LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios." NAACL 2025, long paper.
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. "ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations." NAACL 2025, long paper.
- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. "UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations." ACL 2025, long paper.
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. "Leveraging historical information to boost retrieval-augmented generation in conversations." Information Processing & Management, March 2026.

- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. "Instant Personalized Large Language Model Adaptation via Hypernetwork." pre-print.
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. "POPI: Personalizing LLMs via Optimized Natural Language Preference Inference." pre-print.

Information Extraction

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. "Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition." ACL-IJCNLP 2021, short paper.
- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. "Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition." NAACL 2022, long paper.
- **Pei Chen**, Hang Yang, Kang Liu, et al. "Reconstructing Event Regions for Event Extraction via Graph Attention Networks." AACL-IJCNLP 2020, long paper.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. "Domain-specific Named Entity Recognition via Graph Neural Networks." US Patent 12,530,529, 2026.

Education

Texas A&M University 2019.8 - 2024.5
Ph.D. in Computer Science *College Station, TX*

- **GPA:** 4.0/4.0
- **Research Directions:** Natural language processing, with a focus on information extraction and structured language understanding, including named entity recognition, event extraction, and knowledge base completion; later extended to large language models and agentic systems through research and applied internships.
- **Dissertation:** Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions.

Southwestern University of Finance and Economics 2016.9 – 2018.6
Master's Degree in Finance *Chengdu, China*

- **GPA:** 3.9/5.0, ranking 1/178.
- **Thesis:** Does News Sentiment Predict the Stock Market? An Example on Chinese Growth Market.
- **Awards:** 2017 National Scholarship for Graduate Student.

National University of Defense Technology 2010.9 – 2014.6
B.Engr. in Simulation Engineering *Changsha, China*

- **GPA:** 88.61/100, ranking 1/45.
- **Thesis:** Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.

Professional Service

- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
- 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
- 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
- 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
- 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
- 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

- **LLM & ML:** Distributed Training, Post-training, Supervised Fine-tuning, Preference Optimization, DPO, GRPO, DAPO, Context Management, Personalization, Offline/Online Evaluation, Rubric Design, Long-Context Modeling, Synthetic Data, Multi-Agent Systems Design, RAG, Hallucination Mitigation, Data Flywheel
- **Programming & Frameworks:** Python, PyTorch, Git, Docker, SQL, C/C++, Weights & Biases